1 Baseline Overlapping Generations Model

1.1 The Environment

1.1.1 Consumers and Demographics

People live for two periods: in the first period they are young and in the second they are old. After that, they die.

A young individual supplies inelastically 1 unit of labor and receives a labor income w_t. Old individuals cannot work, so they don't have labor income. Consumers internalize that each unit of savings in period t will be converted to $1 + r_{t+1}$ units in the next period.[1]

Consumers derive utility from consuming and have a life-time utility given by $U = u(c_{Y,t}) + \beta\, u(c_{O,t+1})$, where β is the discount factor, $c_{Y,t}$ (resp. $c_{O,t}$) is the consumption of a young (resp. old) person at time t, and $u(c)$ is the period utility function, which is assumed to be increasing $u'(c) > 0$ and concave $u''(c) < 0$. The typical specification we will use is the logarithmic: $u(c) = \ln c$.

On average, a young person gives rise to n new young people in the next period.[2] That is, letting $N_{Y,t}$, $N_{O,t}$ be the amount of young people and old people, respectively, at time t, and $N_t := N_{Y,t} + N_{O,t}$ be the total amount of people alive at period t.

Remark 1 *We have:*

1. *Given that every young individual becomes old in the next period, we have:* $N_{O,t+1} = N_{Y,t}$.
2. *Given that, on average, a young person gives rise to* n *new young people in the next period, we have:* $N_{Y,t+1} = n \cdot N_{Y,t}$.
3. *Using the first in the second, we have:* $N_{Y,t+1} = n \cdot N_{O,t+1}$ *(also, which is used in the next point:* $N_{Y,t} = n \cdot N_{O,t}$*).*
4. *So, population in the next period is given by:* $N_{t+1} = n \cdot N_t$.
   Proof: $N_{t+1} = N_{Y,t+1} + N_{O,t+1} = n \cdot N_{Y,t} + N_{Y,t} = n \cdot N_{Y,t} + n \cdot N_{O,t} = n \cdot N_t$.[3]

[1] Consumers lend money to firms, who will pay back next period with interests r_{t+1}.

[2] Note that if $n = 1$, then every individual would be substituted by another new individual: there would be constant population. Then, $n > 1$ implies population growth, and $n < 1$ implies that population is declining.

[3] Where in the first and last equality, we use the definition of total population; in the second equality, we use (1) and (2); and in the third one, we use (3).

1.1.2 Firms

There is a representative firm that uses labor L_t and capital K_t to produce Y_t units of a homogeneous final good, according to a neoclassical production function $Y_t = F(K_t, L_t) := AK_t^{\alpha} L_t^{1-\alpha}$.

Labor and capital are supplied by consumers and the final good is sold to consumers.

The final good is the numeraire (that is, the price of the final good is 1, the values are expressed in units of the final good), and the representative firm takes the prices of labor w_t and capital r_t as given, and that a fraction δ of capital gets depreciated.

1.2 Equilibrium

1.2.1 Firm problem

The representative firm chooses the amount of labor L_t and capital K_t to produce a given amount of final output Y_t given its production function $F(L_t, K_t)$ and the price of factors w_t, r_t, in order to maximize profits. That is, the firm solves:

$$\max_{L_t, K_t} F(L_t, K_t) - (r_t + \delta_t) K_t - w_t L_t$$

First order conditions:

$$\begin{aligned} [K_t]: \quad & \alpha \frac{Y_t}{K_t} - (r_t + \delta) = 0, \\ [L_t]: \quad & (1-\alpha) \frac{Y_t}{L_t} - w_t = 0. \end{aligned}$$

The first one implies that capital costs $(r_t + \delta) K_t$ are a fraction α of revenues Y_t. Whereas the second one implies that labor costs $w_t L_t$ are a fraction $1-\alpha$ of revenues Y_t. So, combining them, we have that total costs equal revenues; that is: the representative firm is making zero profits.

1.2.2 Consumers' Problem

Consumers decide how much to consume in each period subject to their budget constraints; that is:

$$\begin{aligned} \max_{c_{Y,t}, c_{O,t+1}} \; & u(c_{Y,t}) + \beta\, u(c_{O,t+1}), \\ \text{s.t. } & w_t = c_{Y,t} + s_{Y,t} && \text{(1st period budget constraint)}, \\ & (1 + r_{t+1}) s_t = c_{O,t+1} + s_{O,t+1} && \text{(2nd period budget constraint)}. \end{aligned}$$

Note that there is no incentive to save in the second period, so old consumers will consume all their savings from the first period: $c_{O,t+1} = (1+r_{t+1})s_t$.[4] Therefore, the choice of savings when young $s_{Y,t}$ determines both the level of consumption when young $c_{Y,t}$ (from the 1st period budget constraint), and the level of consumption when old (from the 2nd period budget constraint plus the result that $s_{O,t+1} = 0$). So, the consumer's problem can be written equivalently as:

$$\max_{s_{Y,t}} u(w_t - s_{Y,t}) + \beta\, u((1+r_{t+1})s_{Y,t}).$$

This captures the trade-off the consumer faces in the first period when deciding how much to save: each unit of present foregone consumption will be converted into $1 + r_{t+1}$ units of future consumption. That is, saving more in the first period decreases consumption (and thus utility) in the first period, but increases consumption (and thus utility) in the second period. The first order condition is:[5]

$$U'(s_{Y,t}) \;=\; -u'(w_t - s_{Y,t}) + \beta\,(1+r_{t+1})\,u'((1+r_{t+1})s_{Y,t}) = 0. \tag{1}$$

This first order condition defines implicitly savings at the young stage $s_{Y,t}$ as a function of factor prices w_t, r_{t+1}. Before getting the explicit solution for the logarithmic utility case, let's analyze how consumers change their savings decision as factor prices change:

- $\frac{\partial s_{Y,t}}{\partial w_t} > 0$ (that is, an increase in wages leads to an increase in savings). Intuition: For a given level of savings $s_{Y,t}$, higher w_t implies higher present consumption; and, since $u(c)$ is concave, this implies that the marginal utility of present consumption decreases, making future consumption relatively more attractive: life-time utility can be increased by increasing savings $s_{Y,t}$.

- The sign of $\frac{\partial s_{Y,t}}{\partial r_{t+1}}$ is ambiguous as there are two conflicting forces:

  - **Substitution effect ($\uparrow r_{t+1} \longrightarrow \uparrow s_{Y,t}$):** Higher r_{t+1} implies that each unit of savings in period t is transformed into more units of future consumption, making saving more attractive (equivalently, present consumption becomes more expensive).

[4] One could get positive savings at the moment of death if either (i) the period of death were uncertain (allowing people to live more than two periods with some probability), or (ii) people cared about leaving a bequest to new generations.

[5] Recall the definition of derivative: $u'(c) := \lim_{\Delta c \to 0} \frac{u(c+\Delta c)-u(c)}{\Delta c}$. That is, $u'(c)$ tells us that for small changes of c, Δc, utility will increase at the rate $u'(c)\Delta c$. Now, for a small change of s, Δs, present consumption changes $\Delta c_t = -\Delta s$, and future consumption by $\Delta c_{t+1} = (1+r_{t+1})\Delta s$; so life-time utility changes by $-\Delta s \cdot u'(c_{Y,t}) + \beta\,(1+r_{t+1})\Delta s \cdot u'(c_{O,t+1})$. If this is positive (resp. negative), the consumer would get a higher utility by choosing $s + \Delta s$ (resp. $s - \Delta s$) instead of s.

– **Income effect ($\uparrow r_{t+1} \longrightarrow \downarrow s_{Y,t}$):** For a given $s_{Y,t}$, higher r_{t+1} implies higher future consumption, and since u is concave, then the marginal utility of future consumption decreases; so they would also prefer to consume more today.

As we will see below, in the logarithmic case, the two effects exactly offset and savings are independent of the interest rate.

1.2.3 Labor market clearing

Labor market clearing states that the wage w_t must be such that labor demand L_t equals labor supply $N_{Y,t}$ (recall that only young individuals supply inelastically one unit of labor):

$$L_t \;=\; N_{Y,t}.$$

1.2.4 Capital market clearing

Let S_t be the aggregate level of savings in the economy at period t. At this point, we know that old people don't save (i.e., $s_{O,t} = 0$) and that young people save $s_{Y,t}$ (recall it is obtained from solving the FOC given in Equation 1, given some values of w_t and r_{t+1}) and there are $N_{Y,t}$ young people; so, we have:

$$S_t = N_{Y,t}\, s_{Y,t}.$$

Consumers lend these savings to firms, which want to borrow K_{t+1} units of final good to buy capital (in equilibrium, the interest rate r_{t+1} must be such that the supply of savings S_t equals the demand for savings K_{t+1}; i.e., $K_{t+1} = S_t$). In the next period, firms (i) use this capital to produce output (i.e., $Y_{t+1} = F(K_{t+1}, L_{t+1})$), (ii) sell the capital at the market price $(1-\delta)S_t$ (its value has depreciated),[6] and (iii) pays back the loan plus interests: $S_t\,(1 + r_{t+1})$.[7] Then, we repeat: firms borrow the aggregate savings at $t + 1$ and use it to buy capital: $K_{t+2} = S_{t+1} = N_{Y,t+1}s_{Y,t+1}$, etc.

The aggregate investment in new capital I_t is

$$I_t \;=\; K_{t+1} - (1-\delta)K_t.$$

That is, if $K_{t+1} > (1-\delta)K_t$ then there is positive investment in period t; whereas if $K_{t+1} < (1-\delta)K_t$, then there is negative investment (disinvestment).

[6] They don't necessarily need to explicitly sell the capital, but this is the value it has at that time, and they could potentially sell it. In particular, if the desired capital for the following period K_{t+2} is lower than $S_t\,(1-\delta)$ (i.e., $K_{t+2} < K_{t+1}\,(1-\delta)$), then the firm has sold capital from last period for a value of at least $K_{t+1}\,(1-\delta) - K_{t+2}$.

[7] Note that the part that it needs to pay from revenues is $S_t\,(1 + r_{t+1}) - (1-\delta)\,S_t = (r_{t+1} + \delta)\,S_t$).

1.2.5 Aggregate resource constraint

Recall the budget constraints of young and old individuals are, respectively:

$$\begin{aligned} w_t &= c_{Y,t} - s_{Y,t}, \\ r_t s_{Y,t-1} &= c_{O,t} + \underbrace{s_{O,t}}_{=0} - s_{Y,t-1}. \end{aligned}$$

Adding up for all the consumers in period t (recall there are $N_{Y,t}$ young individuals and $N_{O,t}$ old ones):

$$N_{Y,t} w_t + N_{O,t} s_{Y,t-1}\, r_t \;=\; N_{Y,t}\,(c_{Y,t} + s_{Y,t}) + N_{O,t}\,(c_{O,t} \;-\; s_{Y,t-1}).$$

Now, we use:

- From the labor market clearing condition: $N_{Y,t} w_t = L_t w_t$.
- From the demographics section, we know that current period old people are the previous period young: $N_{O,t} = N_{Y,t-1}$. And in the capital market clearing section, we have seen that $N_{Y,t-1} s_{Y,t-1} \;=\; S_{t-1} \;=\; K_t$. Therefore: $N_{O,t}\, s_{Y,t-1}\, r_t \;=\; K_t\, r_t$.
- Aggregate consumption C_t is the sum of the consumption made by young and old people; that is: $N_{Y,t}\, c_{Y,t} \;+\; N_{O,t}\, c_{O,t} \;=\; C_t$.
- As in the second point for t and $t-1$ respectively: $N_{Y,t}\, s_{Y,t} \;=\; K_{t+1}$ and $-N_{O,t}\, s_{Y,t-1} \;=\; -K_t$.

So, we have:

$$L_t w_t + K_t\, r_t \;=\; C_t + K_{t+1} \;-\; K_t\,.$$

Next, from the firm problem, we had:

- $L_t\, w_t = (1-\alpha)\, Y_t$;
- $K_t\, r_t = \alpha\, Y_t - K_t\, \delta$.

Adding them, we have $L_t w_t + K_t\, r_t \;=\; (1-\alpha)\, Y_t \;+\; \alpha\, Y_t - K_t\, \delta \;=\; Y_t - K_t\, \delta$. And using this in the previous equation, we have:

$$Y_t \;=\; C_t + K_{t+1} - (1-\delta)\, K_t\,.$$

Finally, using the definition of investment in the capital market clearing subsection, we get the aggregate resource constraint:

$$Y_t \;=\; C_t \;+\; I_t. \tag{2}$$

1.3 The Law of Motion of Capital

Let $k_t := \frac{K_t}{L_t}$ be the capital per capita at period t. We have the following difference equation characterizing the evolution of capital over time (given an initial level of capital K_0):

$$k_{t+1} \;=\; \frac{s_{Y,t}(\,f(k_t)\;-f'(k_t)\,k_t\,,\;f'(k_{t+1})\,)}{n}. \tag{3}$$

Derivation:

- In the consumers' problem section, we have seen that $s_{Y,t}$ is a function of w_t, r_{t+1} (we can state the existence of this relationship explicitly writing $s_{Y,t}(w_t, r_{t+1})$).
- In the capital market clearing section, we have seen that $K_{t+1} \;=\; N_{Y,t}\, s_{Y,t}(w_t, r_{t+1})$.
- Recall from Remark 1 that $N_{Y,t}$ (and from the labor market clearing condition, we know $N_{Y,t} \;=\; L_t$) will be growing at a constant rate over time ($\frac{L_{t+1}}{L_t} \;=\; \frac{N_{Y,t+1}}{N_{Y,t}} \;=\; n$); so, we will try to find a steady state in per capita terms: let $k_t := \frac{K_t}{L_t}$. Therefore: $k_{t+1} := \frac{K_{t+1}}{L_{t+1}} = \frac{N_{Y,t}}{n \cdot L_t}\, s_{Y,t}(w_t, r_{t+1}) \;=\; \frac{s_{Y,t}(w_t, r_{t+1})}{n}$.
- Now, we will use that the neoclassical production function has constant returns to scale; that is, for any scalar $\gamma \geq 0$, it is satisfied that $F(\gamma \cdot K_t, \gamma \cdot L_t) = \gamma \cdot F(K_t, L_t)$. In particular, taking $\gamma = \frac{1}{L_t}$, we have: $\frac{F(K_t,L_t)}{L_t} \;=\; F(\frac{K_t}{L_t}, \frac{L_t}{L_t}) = f(k_t)$.
- If we take the derivative with respect to:
  1. K_t: $\frac{\partial\left(\frac{F(K_t,L_t)}{L_t}\right)}{\partial K_t} \;=\; \frac{\partial F(K_t,L_t)}{\partial K_t}\frac{1}{L_t}$, and $\frac{\partial f(k_t)}{\partial K_t} \;=\; \frac{\partial f(k_t)}{\partial k_t}\frac{1}{L_t}$. So: $\frac{\partial F(K_t,L_t)}{\partial K_t} = f'(k_t)$.
  2. L_t: $\frac{\partial\left(\frac{F(K_t,L_t)}{L_t}\right)}{\partial L_t} \;=\; \frac{\partial F(K_t,L_t)}{\partial L_t}\frac{1}{L_t} - \frac{F(K_t,L_t)}{L_t^2}$, and $\frac{\partial f(k_t)}{\partial L_t} \;=\; -\frac{\partial f(k_t)}{\partial k_t}\frac{k_t}{L_t}$. So: $\frac{\partial F(K_t,L_t)}{\partial L_t} - f(k_t) = -f'(k_t)\,k_t$.
- Using these in the expressions $w_t \;=\; \frac{\partial F(K_t,L_t)}{\partial L_t}$ and $r_{t+1} \;=\; \frac{\partial F(K_{t+1},L_{t+1})}{\partial K_{t+1}} - \delta$ from the firm's problem section, we have:

$$r_{t+1} = f'(k_{t+1}) - \delta\,, \qquad w_t = f(k_t)\;-f'(k_t)\,k_t$$

1.3.1 Steady State

We will say we are in the steady state when capital per capita remains constant thereafter; that is, k^* is a steady state level of capital per capita if $k_t = k^* \implies k_{t+1} = k^*$, which is the case if and only if Equation 3 is satisfied with $k_t = k^*$, $k_{t+1} = k^*$:

$$k^* \;=\; \frac{s_{Y,t}(\,f(k^*)\;-f'(k_t)\,k^*\,,\;f'(k^*)\,)}{n}. \tag{4}$$

There are some natural questions to ask:

1. **Does a steady state exist? Is it unique, or how many are there?**

2. **Do we necessarily converge to it? Is it locally convergent (in which case, we would say it is a stable steady state)?** By convergence to k^*, we mean that, starting from some k_0, as time goes to infinity, we move closer and closer to k^*. By local convergence to k^*, we mean that if we start with k_0 sufficiently close to k^*, then we converge to k^*.

There is always a trivial steady state with zero capital. We will typically discard this degenerate steady state, and focus with steady states with positive capital.
Without further assumptions on the utility function, it is difficult to say much about these questions: the savings function could be such that there does not exist any steady state, or such that there exist more than one. We will analyze these questions in the particular cases below.

Efficiency: Here, we ask: is the steady state equilibrium of the OLG model efficient? Short answer: not necessarily.
To assess this, we would compare welfare under the steady state of the competitive equilibrium (i.e., the previous equilibrium) with the social optimal (i.e., the welfare that would be obtained if a social planner were to make the decisions).
Here, compared to a model with a representative consumer, a problem arises: how should we weight the welfare of different generations? To avoid having to resolve this issue, we can restrict our focus on **Pareto optimality**. Recall that an equilibrium is Pareto optimal if you cannot improve someone's utility without harming another person.
The social planner would choose young and old consumption to maximize welfare (as mentioned above, not clear how to define it with different consumers), subject to the aggregate resource constraint $Y_t = C_t + K_{t+1} - (1-\delta)K_t$. Since our focus is in the steady state in per capita terms, divide this constraint by L_t, to get:

$$f(k) = c + n\,k - (1-\delta)\,k,$$

where $c := \frac{C_t}{L_t} = c_Y + \frac{c_O}{n}$, where c_Y and c_O are the steady state levels of consumption of a young individual and an old individual, respectively. Note that the planner could solve this in two steps: first, choose aggregate savings per capita (i.e., k) such that consumption per capita c is maximized, and then allocate the aggregate consumption between young and old consumers. That is, in the first step, the planner solves:

$$\max_k c(k) = Ak^\alpha + k(1 - \delta - n)$$

Note that this first step refers to Pareto optimality: if $c(k_1) < c(k_2)$, then k_2 is a Pareto improvement over k_1 (for instance, if you increase k from k_1 to k_2, you could maintain the consumption of young and increase the consumption of old).
Now, Given that $c'(k) = \alpha \frac{f(k)}{k} + (1 - \delta - n)$:

- If $n < 1$, we have a degenerate steady state with no population (recall n was defined as the number of new young people generated from each young person; so $n < 1$ means population declines sustainedly).

- If $n > 1$, then the first-order condition says that the planner will choose k^S such that

  $$\alpha\, A(k^S)^{\alpha-1} = n - (1 - \delta)\,(> 0).$$

  That is, such that the marginal product of capital per capita equals the sum of the population growth rate and the depreciation rate (that is, the marginal units of capital per capita that are just enough to compensate for the depreciation of capital and to keep up with population growth).
  Solving for k^S (golden rule capital per capita):

  $$k^S = \left(\frac{\alpha\, A}{n - 1 + \delta}\right)^{\frac{1}{1-\alpha}}$$

  So, we have to compare k^S with k^* from Equation 7. The competitive equilibrium is only efficient if $k^S = k^*$; otherwise there is some inefficiency: consumers may over-accumulate capital $k^S < k^*$ or under-accumulate capital $k^S > k^*$.

  In the case of over-accumulation, you could actually cut savings today and increase both consumption today (given some output, you are saving less and consuming more), as well as consumption tomorrow (as the lower capital stock causes a reduction in replacement costs that dominates the reduction in production).

  In the case of under-accumulation, although in the long run consumption is higher if you increase capital, this comes at a short-run cost (in order to accumulate more capital, you need to save more today, which means you need to cut present consumption).

  There is over-accumulation (that is, the social planner would want consumers to save less) if and only if:

$$\frac{\alpha}{n-1+\delta} < \frac{\beta}{1+\beta}\frac{(1-\alpha)}{n}.$$

This is more likely to be the case when:

- α is close to 0. The social planner internalizes that increasing k today will allow more production tomorrow. When α is close to 0, this effect is almost negligible (regardless of the level of k, production will be almost the same), and so the social planner sees increasing k as a waste of resources.
- The discount factor β is closer to 1 (i.e., the higher the value placed on future utility). The consumer saves when young because they value utility (from consumption) when old. The higher β, the greater the weight the consumer places on old-age consumption, and thus the more they want to save.
- The larger the difference $n - (1-\delta)$ (≥ 0). On the one hand, the higher the depreciation rate δ, the more likely the inequality holds. If $\delta = 1$, then capital accumulated today is only used for production for one period, after which it loses all its value. Thus, the planner internalizes that the social value of accumulating capital to increase future production is low. On the other hand, the planner internalizes that the marginal cost of increasing capital per capita increases with population growth (a higher n implies that more resources are needed to maintain a given level of capital per capita). Note that if $n = 1-\delta$, then the planner internalizes that it can keep any given level of capital per capita without saving (that is, the marginal cost of replacing capital per capita is 0, and so the planner would choose to accumulate capital per capita indefinitely).[8]

1.4 Particular specifications of $u(c)$

1.4.1 Logarithmic preferences

Consumer problem: If $u(c) := \ln c$, then Equation 1 reads:

$$-\frac{1}{w_t - s_{Y,t}} + \beta(1+r_{t+1})\frac{1}{(1+r_{t+1})s_{Y,t}} = 0.$$

[8] To see this, consider a given level of capital and no savings; then, capital decreases over time, but at the same rate at which the population is declining.

On the one hand, we can rewrite it using $c_{Y,t} = w_t - s_t$ and $c_{O,t+1} = s_{Y,t}(1 + r_{t+1})$ (budget constraints) to obtain the Euler equation (which characterizes the evolution of consumption):

$$\frac{c_{O,t+1}}{c_{Y,t}} = \beta(1 + r_{t+1}). \tag{5}$$

On the other hand, we can solve for $s_{Y,t}$:

$$-\frac{1}{w_t - s_{Y,t}} + \beta(1 + r_{t+1})\frac{1}{(1 + r_{t+1})s_{Y,t}} = 0 \implies 1 = \beta\frac{w_t - s_{Y,t}}{s_{Y,t}} \implies s_{Y,t} = w_t\frac{\beta}{1+\beta}.$$

Then, from the 1st period budget constraint $c_{Y,t} = w_t - s_t$, we have consumption in the young stage:

$$c_{Y,t} = w_t\frac{1}{1+\beta}.$$

And then, using the Euler equation, we get the consumption at the old stage:[9]

$$c_{O,t+1} = w_t(1 + r_{t+1})\frac{\beta}{1+\beta}.$$

Law of motion of capital: Using the savings function we have found: $s_{Y,t}(w_t) = w_t\frac{\beta}{1+\beta}$, together with the expression of wage from the firm's problem, $w_t = (1-\alpha)\frac{Y_t}{L_t} = (1-\alpha)A\,k_t^{\alpha}$; then, Equation 3 becomes:

$$k_{t+1} = k_{t+1}(k_t) := \frac{\beta}{1+\beta}\frac{(1-\alpha)A}{n}k_t^{\alpha}. \tag{6}$$

For convenience, we can define $\gamma := \frac{\beta}{1+\beta}\frac{(1-\alpha)A}{n}$; so that $k_{t+1}(k) = \gamma k^{\alpha}$.

Now, is there a steady state? If so, how many? And do we necessarily converge to this steady state?
We can directly solve for $k^* > 0$ such that $k^* = \gamma\, {k^*}^{\alpha}$:

$$k^* = \gamma^{\frac{1}{1-\alpha}}. \tag{7}$$

So, by directly finding the steady state level of capital per capita, we have proved the existence of a steady state, and its uniqueness. Next, we will show that k_t monotonically converges to k^*, regardless of the initial k_0 (for a graphical proof, see Figure 1):
Proof. We need to prove that (i) $k_t < k^* \implies k_t < k_{t+1}(k_t) \leq k^*$ and (ii) $k_t > k^* \implies k_t > k_{t+1}(k_t) \geq k^*$.

[9]Equivalently, you can get it using the solution for $s_{Y,t}$ and the 2nd period budget constraint $c_{O,t+1} = s_{Y,t}(1 + r_{t+1})$.

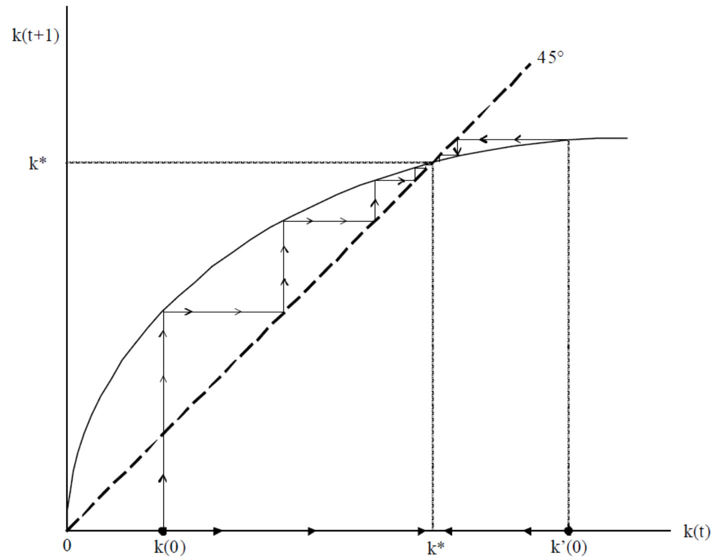


Figure 1: Illustration of the monotone convergence and dynamics of k_t in the basic log utility and Cobb-Douglas production function OLG model.

1. First, $k'_{t+1}(k_t) = \alpha\,\gamma\,k_t^{\alpha-1} > 0$, together with $k^* = k_{t+1}(k^*)$ implies: $k_t < k^* \implies k_{t+1}(k_t) < k^*$, and $k_t > k^* \implies k_{t+1}(k_t) > k^*$. So, we are only left to prove: (i) $k_t < k^* \implies k_t < k_{t+1}(k_t)$ and (ii) $k_t > k^* \implies k_t > k_{t+1}(k_t)$.

2. Note that for $k_t = 0$: $k_t = 0 = k_{t+1}(k_t)$, and $\lim\limits_{k\to 0} k'_{t+1}(k) = \infty > 1 = \frac{\partial k_t}{\partial k_t}$, which implies that for $\epsilon > 0$ sufficiently small we have $k_t < k_{t+1}(k_t)$ for any $k_t \in (0, \epsilon]$.

3. Given that we already know there is only one steady state, and since the function $k_{t+1}(k_t) - k_t$ is continuous, then $k_t < k_{t+1}(k_t)$ for any $k_t < k^*$.

4. Note that $k'_{t+1}(k^*) < 1$: By contradiction, suppose $1 \le k'_{t+1}(k^*) = \lim\limits_{\delta\to 0} \frac{k_{t+1}(k^*) - k_{t+1}(k^*-\delta)}{\delta}$, which implies $k^* = k_{t+1}(k^*) \ge k_{t+1}(k^* - \delta) + \delta \iff k^* - \delta \ge k_{t+1}(k^* - \delta)$. But from the previous point, given that $k^* - \delta < k^*$, we have the contradiction: $k^* - \delta \ge k_{t+1}(k^* - \delta) > k^* - \delta$.

5. Finally, given that $k''_{t+1}(k_t) = -(1-\alpha)\frac{k'_{t+1}(k_t)}{k_t} < 0$ and in the previous point we have seen that $k'_{t+1}(k^*) < 1$; then $k'_{t+1}(k_t) < 1$ for all $k_t \ge k^*$, and so: $k_{t+1}(k_t) < k_{t+1}(k^*) + (k_t - k^*) = k_t$, for any $k_t > k^*$, as wanted.

■

1.4.2 Constant Intertemporal Elasticity of Substitution utility function

If $u(c) := \frac{c^{1-\theta}-1}{1-\theta}$, which is a Constant Intertemporal Elasticity of Substitution (CIES) utility function (below, we will see why), then Equation 1 reads:

$$-\left(w_t - s_{Y,t}\right)^{-\theta} + \beta(1+r_{t+1})\left(s_{Y,t}(1+r_{t+1})\right)^{-\theta} = 0. \tag{8}$$

On the one hand, we can solve as before for $s_{Y,t}$, and now we get:

$$(w_t - s_{Y,t})^{-\theta} = \beta(1+r_{t+1})^{1-\theta} s_{Y,t}^{-\theta} \implies \frac{1}{\beta(1+r_{t+1})^{1-\theta}} = \left(\frac{w_t}{s_{Y,t}} - 1\right)^{\theta} \implies s_{Y,t} = w_t \frac{\beta^{\frac{1}{\theta}}(1+r_{t+1})^{\frac{1-\theta}{\theta}}}{1+\beta^{\frac{1}{\theta}}(1+r_{t+1})^{\frac{1-\theta}{\theta}}}.$$

As before, using the budget constraints, you can get the levels of young and old consumption. Note that for $\theta = 1$, we have exactly the same savings function as in the log case (in fact, as shown in the footnote, the log utility is a particular case of the CIES utility with $\theta = 1$).[10] You can also check that when $\theta < 1$ (resp. $\theta > 1$), then $s_{Y,t}$ is increasing (resp. decreasing) in r_{t+1}; that is, the substitution (resp. income) effect dominates.

On the other hand, using $c_{Y,t} = w_t - s_t$ and $c_{O,t+1} = s_{Y,t}(1+r_{t+1})$ (budget constraints), we can obtain the Euler equation:

$$\frac{c_{O,t+1}}{c_{Y,t}} = \left(\beta(1+r_{t+1})\right)^{\frac{1}{\theta}}. \tag{9}$$

We define the Intertemporal Elasticity of Substitution (IES) of an individual $\sigma := \frac{\partial \frac{c_{O,t+1}}{c_{Y,t}}}{\partial r_{t+1}}$ as how much the growth of consumption of the individual changes when the interest rate increases. From the Euler equation above, it is straightforward that in the CIES utility case, we have $\sigma = \frac{1}{\theta}$. So:

1. $\theta < 1$ implies $IES > 1$ and so the substitution effect dominates and an increase in r_{t+1} induces the consumer to consume less today and save more, because future consumption is now relatively cheaper (more rewarding).

2. $\theta > 1$ implies $IES < 1$ and so the income effect dominates and an increase in r_{t+1} induces the consumer to save less, thereby increasing not only future consumption but also present consumption.

[10] Note that $\lim_{\theta \to 1} \frac{c^{1-\theta}-1}{1-\theta} = \lim_{h \to 0} \frac{c^h - 1}{h} = \frac{\partial c^x}{\partial x}\Big|_{x=0} = \ln c \cdot e^0 = \ln c.$

1.5 Exercises:

Exercise 1: Consider the OLG model discussed in class with logarithmic utility and the following Cobb-Douglas function $Y_t = AK_t^{\alpha}L_t^{1-\alpha}$. The household works for a wage w_t when young and has no labor income when old, but can save (r_t is the gross interest rate). The firm is competitive.

1. Obtain the optimal consumption path and saving function for the household.
2. Derive how the per capita capital stock k_{t+1} tomorrow depends on the per capita capital stock k_t today.
3. Explain the impact of an increase in n in k_{t+1}.
4. Find the steady state capital stock(s) in the economy. How many steady states are there?

Exercise 2: Consider the OLG model discussed in class with logarithmic utility and Cobb-Douglas production function $Y_t = BK_t^{\alpha}(A_tL_t)^{1-\alpha}$. Markets are competitive (labor and capital earn their marginal products and firms earn zero profits). The young divide their labor income w_tA_t between consumption and saving (w_t represents wage per unit of effective labor).

1. Write down the household problem (the utility and budget constraints).
2. Obtain the saving function for the household.
3. Let k_t be the capital per unit of effective labor. Obtain the equation for k_{t+1} as a function of k_t.
4. Describe how k_{t+1} is affected by a reduction in B.
5. Describe how k_{t+1} is affected by an increase in α.

2 Pension Systems

2.1 Introduction

The increase in life expectancy coupled with the decrease in birth rates have cause a rise in the ratio between retired population and working population, putting pressure on the sustainability of the pension systems of many countries.

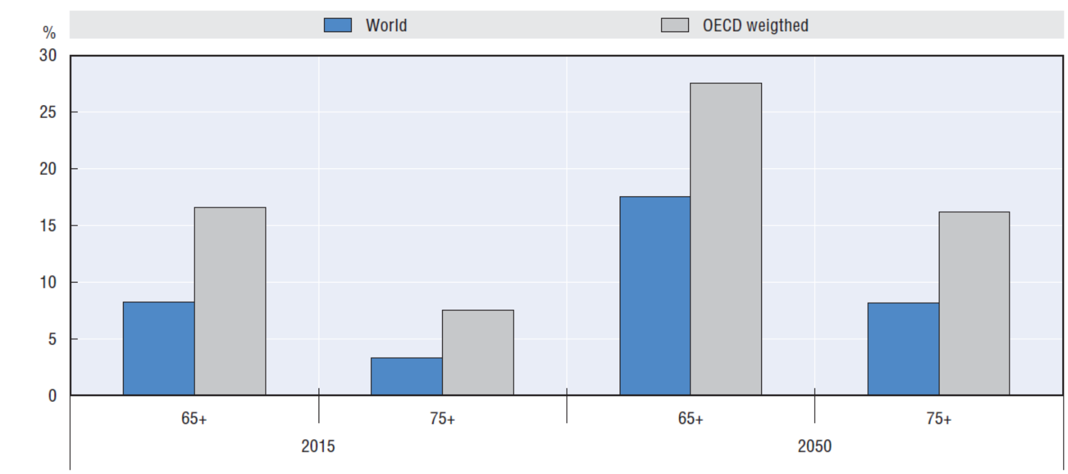
First, we will introduce some concepts, and then we will extend the previous OLG model to incorporate pensions.

- **Total fertility rate:** Number of children each woman would have if she lived to the end of her fertile life and if the probability of giving birth at each age equals the average fertility rate for that age. That is, let $b_{a,t} := \frac{\text{number of births from women of age a in year t}}{\text{number of women with age a in year t}}$ be the average number of births per woman of age a in year t. Then, we define the total fertility rate as $\sum\limits_{a=a_{min}}^{a_{max}} b_{a,t}$.
- **Life expectancy:** Average number of years that people of a certain age can expect to live if they experienced the age-specific mortality rates prevailing in a given country at a specific time. That is, let $m_{a,t} := \frac{\text{number of deaths of people with age a in year t}}{\text{number of people with age a in year t}}$ be the average number of people with age a that die in year t. If we take the probability of dying at age a to be $m_{a,t}$, then the probability of living more than a years is $P\{\text{living more than a years}\} = \prod\limits_{i=0}^{a}(1 - m_{i,t})$.[11] Then, we define the life expectancy at birth as $\sum\limits_{a=1}^{a_{max}} \prod\limits_{i=0}^{a}(1 - m_{i,t})$.[12] For a person that is already (for instance) 30 years old, their life expectancy would be $30 + \sum\limits_{a=30}^{a_{max}} \prod\limits_{i=0}^{a}(1 - m_{i,t})$.
- **Dependency ratio:** Working-age population (20-64 years) divided by retirement-age population (65+).
- **Gross replacement rate:** Gross pension payment divided by gross payment before retirement. Clarification: we mean annual earnings, not cumulative earnings. Typically, for gross payment before retirement, we take the last annual gross earning before retirement, or the average over some period before retirement.

[11] We can prove this by induction on a. For $a = 0$, it is clearly true: $P\{\text{living more than 0 years}\} = 1 - m_{0,t}$. Suppose it is true for a, and check for $a + 1$: $P\{\text{living more than } a+1 \text{ years}\} = P\{\text{living more than a years}\} \cdot P\{\text{not dying at age } a+1\} = \left(\prod\limits_{i=0}^{a}(1 - m_{i,t})\right)(1 - m_{a+1,t}) = \prod\limits_{i=0}^{a+1}(1 - m_{i,t})$, which completes the proof.

[12] Note that the expected value of a discrete variable x that can take the values $\{1, 2, \ldots, n\}$ is $\mathbb{E}[x] = \sum\limits_{i=1}^{n} iP\{x = i\} = \sum\limits_{k=1}^{n}\sum\limits_{i=k}^{n} P\{x = i\} = \sum\limits_{k=1}^{n} P\{x \geq k\}$.

Figure 1.1. **Share of elderly older than 65 and 75 in the total population**



Source: United Nations (2013), *World Population Prospects: The 2012 Revision* and OECD calculations.

Figure 2: Percentage of elderly in the population. We get two observations from this figure: 1) OECD countries have a higher share of elderly people; and 2) the share of elderly people is expected to increase.

Table 7.1. Total fertility rates, 1960-2060

	1960	1970	1980	1990	2000	2010	2013	2020*	2060*
OECD members									
Australia	3.45	2.86	1.89	1.90	1.76	1.95	1.88	1.87	1.86
Austria	2.69	2.29	1.65	1.46	1.36	1.44	1.44	1.57	1.79
Belgium	2.54	2.25	1.68	1.62	1.64	1.84	1.76	1.89	1.95
Canada	3.90	2.33	1.68	1.71	1.49	1.63	1.61	1.74	1.86
Finland	2.71	1.83	1.63	1.79	1.73	1.87	1.75	1.87	1.90
France	2.74	2.48	1.95	1.78	1.87	2.02	1.98	1.98	1.99
Germany	2.37	2.03	1.56	1.45	1.38	1.39	1.41	1.50	1.69
Greece	2.23	2.40	2.23	1.40	1.27	1.47	1.30	1.61	1.80
Italy	2.41	2.43	1.68	1.36	1.26	1.41	1.39	1.61	1.83
Japan	2.00	2.13	1.75	1.54	1.36	1.39	1.43	1.54	1.78
Korea	6.00	4.53	2.82	1.57	1.47	1.23	1.19	1.46	1.75
Mexico	6.78	6.72	4.71	3.36	2.65	2.28	2.22	1.94	1.76
Netherlands	3.12	2.57	1.60	1.62	1.72	1.80	1.68	1.81	1.88
Portugal	3.10	2.83	2.18	1.56	1.56	1.39	1.21	1.38	1.71
Spain	2.86	2.90	2.22	1.36	1.23	1.37	1.27	1.63	1.83
Sweden	2.20	1.94	1.68	2.14	1.55	1.98	1.89	1.95	1.99
Switzerland	2.44	2.10	1.55	1.59	1.50	1.54	1.52	1.62	1.80
Turkey	6.40	5.00	4.63	3.07	2.27	2.06	2.07	1.89	1.77
United Kingdom	2.72	2.43	1.90	1.83	1.64	1.92	1.83	1.89	1.90
United States	3.65	2.48	1.84	2.08	2.06	1.93	1.86	1.98	1.99
OECD34	**3.18**	**2.73**	**2.17**	**1.91**	**1.67**	**1.75**	**1.67**	**1.77**	**1.85**
Other major economies									
Brazil	6.21	5.02	4.07	2.81	2.36	1.84	1.81	1.71	1.75
China	5.76	5.47	2.71	2.51	1.51	1.65	1.66	1.72	1.84
India	5.87	5.49	4.68	3.88	3.15	2.56	2.51	2.25	1.85
Russian Federation	2.42	2.01	1.90	1.89	1.20	1.57	1.69	1.66	1.85
EU28	2.75	2.43	2.01	1.78	1.48	1.59	1.56	1.68	1.84

Source: OECD (2014), Society at a Glance 2014: OECD Social Indicators; United Nations, World Population Prospects.

Figure 3: Fertility rates. We see a generalized reduction in fertility rates.

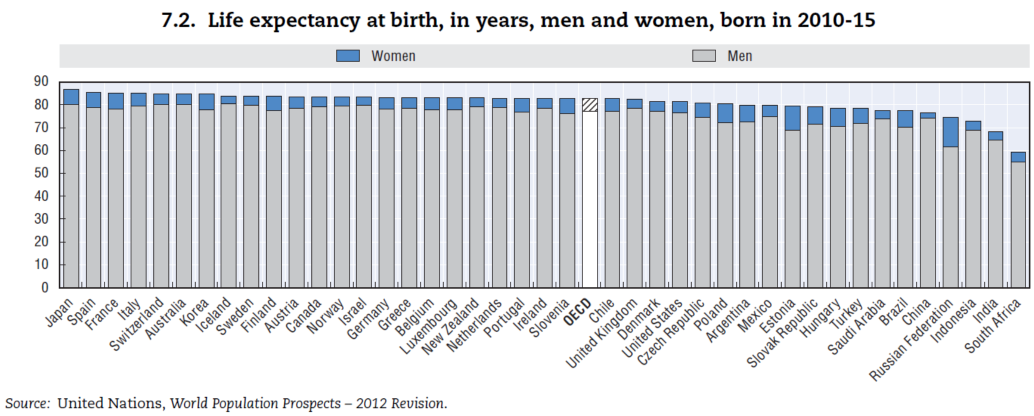
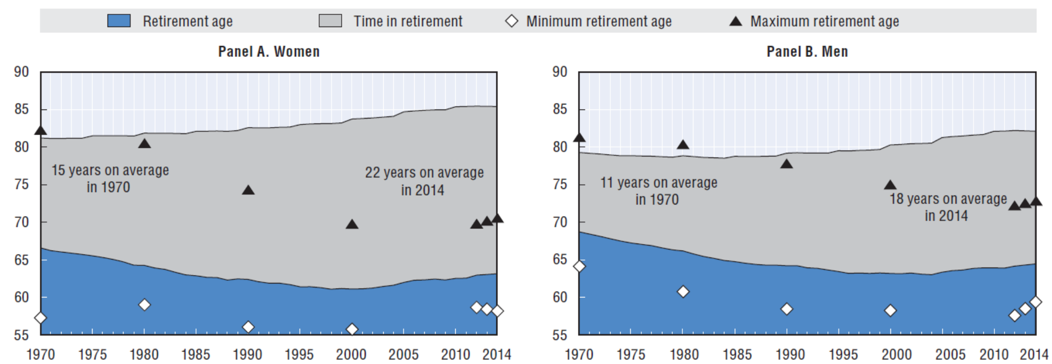


Figure 4: Life expectancy. We see that it is a generalized result across countries that women are expected to live longer.

7.11. Average years in retirement across all OECD countries, 1970-2014



Source: OECD estimates based on the results of national labour force surveys, the European Union Labour Force Survey (EU-LFS) and, for earlier years in some countries, national censuses.

Figure 5: Number of years in retirement. We observe: 1) there has been a decline in the average retirement age prior to 2000, which together with the increase in life expectancy, explains the 2) increase in the average time in retirement.

Table 7.5. Demographic old-age dependency ratios: Historical and projected values, 1950-2075

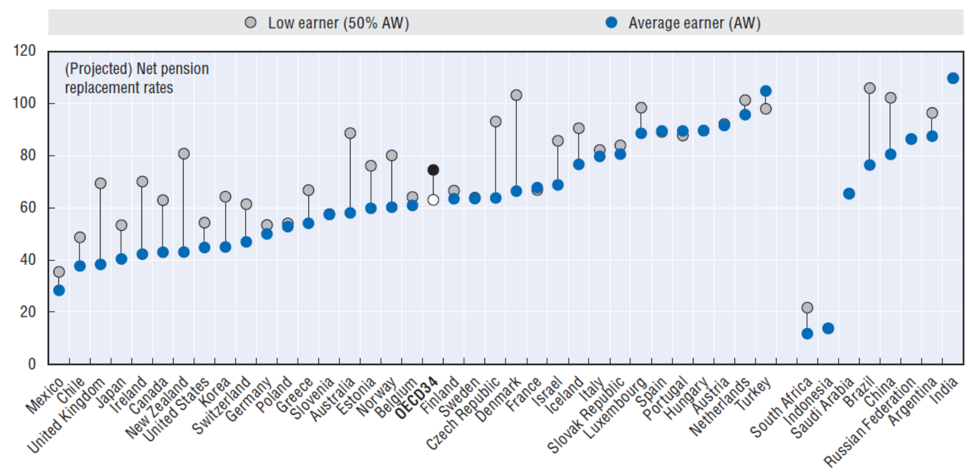
	1950	1975	2000	2015	2025	2050	2075
OECD members							
Australia	14.0	16.0	20.7	25.1	31.5	40.8	47.3
Austria	17.1	27.3	25.3	30.3	36.5	52.8	53.8
Belgium	18.1	25.2	28.4	31.7	39.0	50.5	51.6
Canada	14.0	15.3	20.4	25.9	35.7	46.4	49.9
Finland	11.9	18.0	24.7	35.4	44.6	48.9	53.1
France	19.5	24.6	27.5	32.8	39.7	49.0	53.4
Germany	16.0	26.3	26.2	35.3	43.7	65.1	66.3
Greece	12.5	21.8	27.5	33.5	40.2	65.3	59.7
Italy	14.3	21.7	29.4	36.5	42.6	68.3	63.3
Japan	10.0	13.0	27.6	47.2	55.4	78.4	77.2
Korea	6.3	7.5	11.5	19.6	31.1	71.5	80.1
Mexico	7.9	9.5	9.7	12.1	16.2	35.3	58.7
Netherlands	14.0	19.5	21.9	30.5	39.5	52.5	55.9
Portugal	13.0	19.7	26.7	31.7	38.5	69.8	75.6
Spain	12.8	19.2	27.3	29.6	36.3	73.2	65.4
Sweden	16.8	26.2	29.5	34.8	39.0	42.7	46.5
Switzerland	15.8	21.5	24.9	29.5	34.0	44.3	51.7
Turkey	6.5	10.1	11.3	13.1	17.6	37.4	55.2
United Kingdom	17.9	25.4	26.8	30.8	35.5	46.4	51.0
United States	14.3	19.1	20.9	24.7	33.1	39.5	45.0
OECD (unweighted)	**13.7**	**19.4**	**22.5**	**27.6**	**34.6**	**51.0**	**55.4**
OECD (weigthed)	**13.9**	**18.7**	**21.9**	**27.3**	**34.1**	**48.5**	**54.5**
Major economies							
Brazil	6.5	8.9	10.1	13.3	18.3	39.6	60.2
China	8.7	9.7	11.6	14.2	21.3	42.5	50.6
India	6.4	7.6	8.5	9.6	12.2	20.8	33.3
Russian Federation	8.7	15.4	20.3	20.1	27.7	36.0	34.5
EU28 (unweighted)	14.8	21.1	24.3	29.3	36.3	52.9	55.2

Note: The demographic old-age dependency ratio is defined as the number of individuals aged 65 and over per 100 people of working age defined as those aged between 20 and 64.

Source: United Nations, World Population Prospects – 2012 Revision.

Figure 6: Dependency rate. We observe a generalized increase in dependency rates across countries (and it is expected to continue growing).

Figure 1.7. **Future net replacement rates for low and average income earners in OECD and G20 countries**



Note: The net replacement rate is calculated assuming labour market entry at age 20 in 2014 and a working life equal to the pensionable age in each country The net replacement rates shown are calculated for an individual with 100% and 50% of average worker earnings (AW).

Figure 7: Expected replacement rate in the future across different countries.

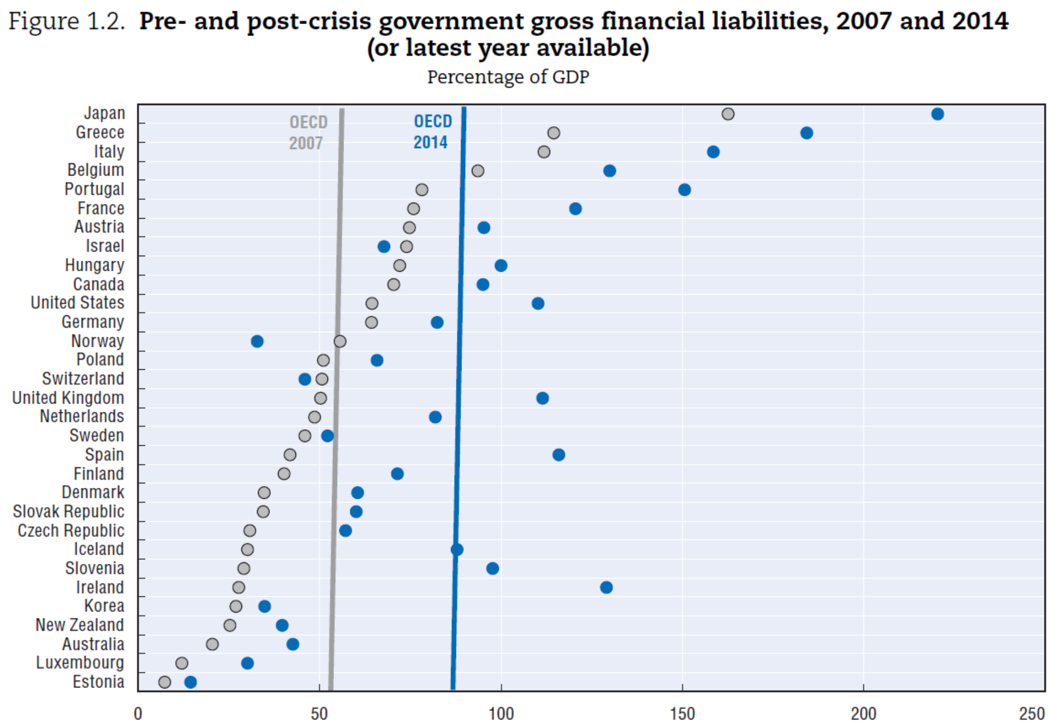


Figure 8: Public Debt. With very few exceptions, public debt has increased in this period.

Figure 1.5. **Current and future retirement ages for a man entering the labour market at age 20**

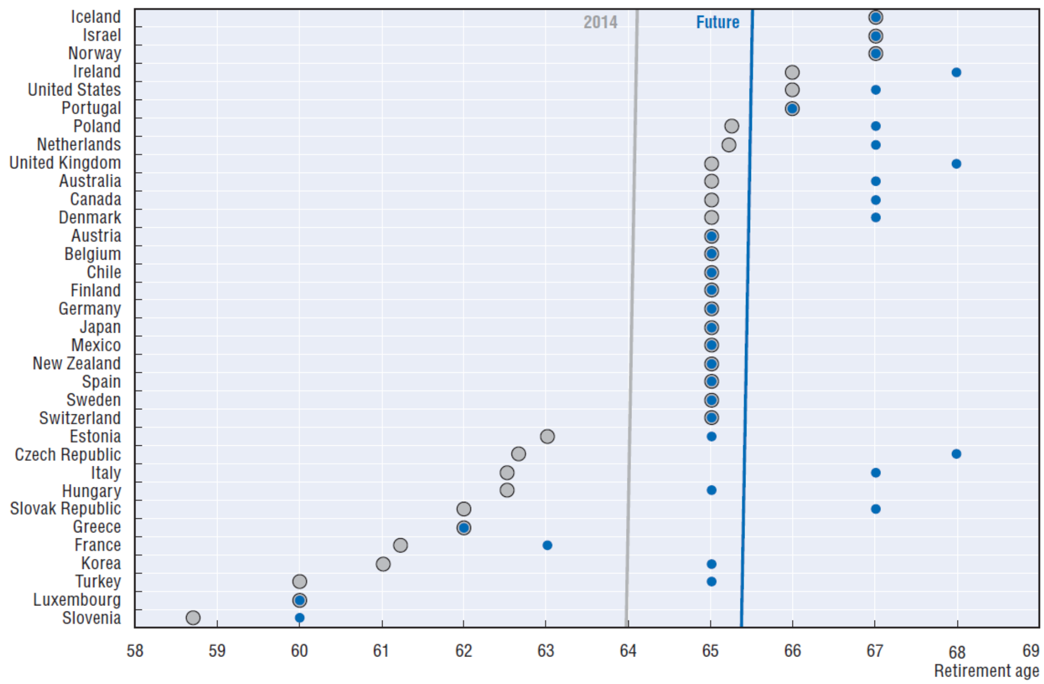


Figure 9: Retirement age. Retirement ages are expected to rise.

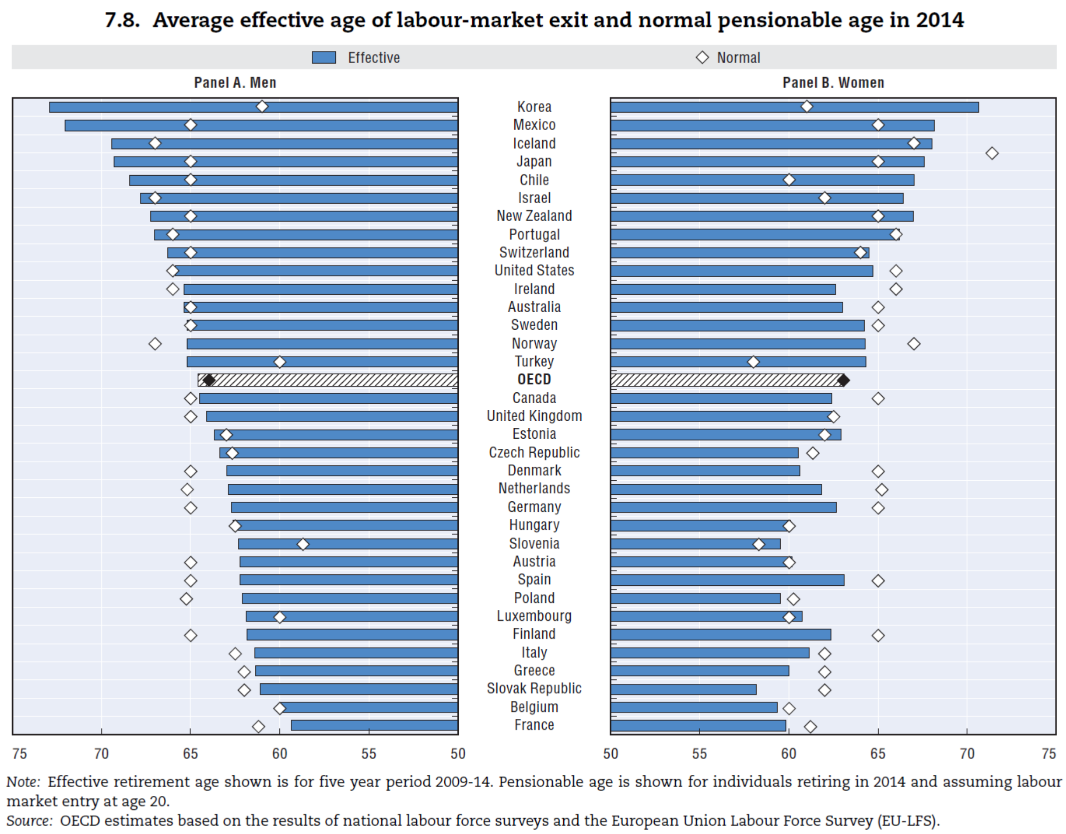
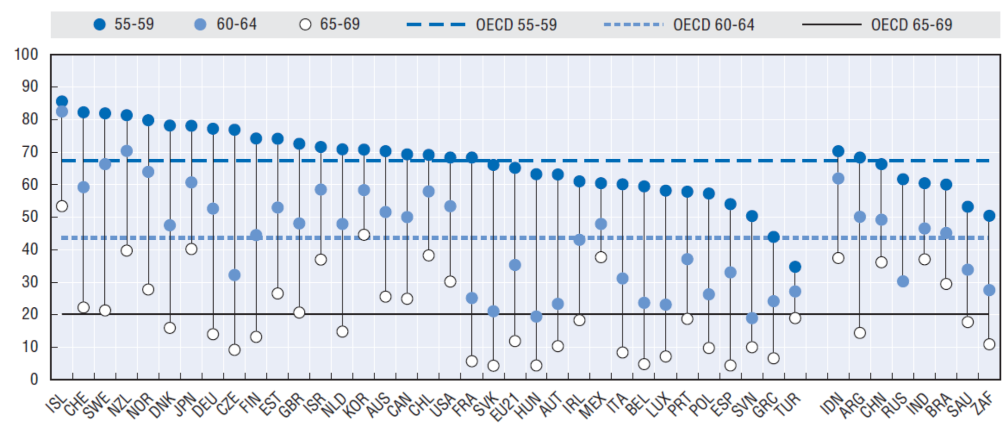


Figure 10: Effective retirement age. Here, normal pensionable age, refers to the official/legal age at which someone (in normal conditions) is eligible for a full public pension. We see that in many European countries, people typically retire before they officially have to.

Figure 1.4. **Employment rate for people aged 55-59, 60-64 and 65-69, OECD and G20 countries, 2014**



Note: Employment rates for non-OECD G20 countries are latest available.
Source: OECD (2015), *OECD Employment Outlook 2015*, OECD Publishing, Paris, *http://dx.doi.org/10.1787/empl_outlook-2015-en*.

Figure 11: Employment rates decrease as people get closer to the retirement age.

2.2 Introducing pensions to the OLG model

Given that we have seen that the competitive equilibrium may not be efficient, there is a role for the government to implement policies that correct this inefficiency.

Consider the following government budget constraint:

$$\underbrace{s_t}_{\text{Surplus}} = \underbrace{\underbrace{N_{Y,t}\, w_t\, \tau_{L,t}}_{\text{Revenue}} - \underbrace{N_{O,t}\, e_t}_{\text{pension payments}} - \underbrace{G_t}_{\text{Other expenses}}}_{\text{Primary surplus}} - \underbrace{r_t\, D_t}_{\text{Debt servicing}} \tag{10}$$

where $\tau_{L,t}$ is the tax rate on labor income, e_t is the amount paid as a pension, G_t are other government expenses not included in pensions, r_t is the interest paid on public debt, and D_t is the public debt at time t, which evolves according to

$$D_{t+1} = D_t - s_t.$$

Observation: Note that if at each period the government borrows b from each young consumer, then in the next period: it has to pay back the principle plus interests $(1+r_{t+1})\, N_{Y,t}\, b$ and it receives (the borrowing from the next generation of young people) $N_{Y,t+1}\, b = (1+n)\, N_{Y,t}\, b$; so, as long as $n \geq r_{t+1}$, then the government can issue debt with no cost (it can sustainedly repeat this strategy); that is, in this case the system is self-financing.

Note that a reduction in the dependency ratio (that is, a decrease in the number of young people per old person, $\downarrow \frac{N_{Y,t}}{N_{O,t}}$) puts greater pressure on the government budget. The government has several policy options to address this situation:

- Increase taxes ($\uparrow \tau_{L,t}$);
- Reduce pension benefits ($\downarrow e_t$);
- Increase the dependency ratio ($\uparrow \frac{N_{Y,t}}{N_{O,t}}$). This can be achieved in various ways:
  - Increase the retirement age so that people work for more years (some individuals who would be counted in $N_{O,t}$ are instead counted in $N_{Y,t}$).
  - Implement policies that facilitate immigration (immigrants are typically of working age).
  - Increase incentives to have children.
- Increase wages ($\uparrow w_t$) through policies that promote productivity growth.
- Cut other expenses ($\downarrow G_t$).

We are going to consider two pension systems:

- **Pay-as-you-go system (PAYG):** Pensions are financed directly with taxes from the current working population.

- **Funded system:** Workers' contributions are invested in the financial system, and pensions are paid with the returns on these investments. If we assume that the returns the government can obtain from the financial system are the same consumers would obtain from their savings, then you can equivalently think of this system as the government forcing consumers to save. Note that if consumers are rational and the amount consumers are forced to save is not too large, then no actual effect on capital accumulation (that is, if the government forces you to save at least $\bar{s}$, but you already wanted to save a larger amount $s > \bar{s}$, then the constraint is not binding and your behavior won't change).[13]

In the following, we are going to develop the PAYG system and leave the fully funded system as an exercise.

2.2.1 Pay-as-you-go system

The government: We assume the government: (i) imposes a proportional tax on labor income $\tau_{L,t}$, (ii) pays a pension e_t to each old person, and (iii) runs a balanced budget (in the sense that revenues from the tax on workers equals the expenses from paying the pensions to the elderly: $N_{Y,t} w_t (1 + \tau_{L,t}) = N_{O,t} e_t$).

Dividing the balanced budget condition by $N_{Y,t-1}$:

$$\frac{N_{Y,t}}{N_{Y,t-1}} \, w_t \, \tau_{L,t} = \frac{N_{O,t}}{N_{Y,t-1}} \, e_t \, .$$

Letting n_t be the number of young people in period t per each young person in the previous period $t-1$ (before we were assuming constant n, now we allow it to change); then, the balanced budget condition reads (using that $N_{Y,t} = n_t \, N_{Y,t-1}$ and $N_{O,t} = N_{Y,t-1}$):

$$n_t \, w_t \, \tau_{L,t} = \; e_t \, . \tag{11}$$

[13]In an extended version of the OLG model with multiple periods of life (rather than only young and old), an additional nuance arises. Forced savings are typically illiquid until retirement, so the relevant comparison is between $\bar{s}$ and the amount the individual would choose to allocate to retirement savings. In this case, a consumer may save more than $\bar{s}$ in total, while still being constrained, because she also wishes to hold liquid savings (available for consumption) before retirement.

Note that a higher population growth (i.e., higher n_t) allows the government to increase pensions (i.e., $\uparrow e_t$), or maintain the same level of pensions with a lower labor tax on the young (i.e., $\downarrow \tau_{L,t}$).

The consumers' problem: The only change in the consumers' problem is in the budget constraint: (i) in the young period, now consumer's income is $w_t\,(1-\tau_{L,t})$, and (ii) in the old period, apart from the returns on savings, now the old person has a new source of income: pension e_t. So, now the problem reads:

$$\max_{s_{Y,t}} u(w_t\,(1-\tau_{L,t}) - s_{Y,t}) + \beta\, u((1+r_{t+1})s_{Y,t} + e_{t+1}).$$

And the first order condition is:

$$-u'(w_t\,(1-\tau_{L,t}) - s_{Y,t}) + \beta\,(1+r_{t+1})\,u'((1+r_{t+1})s_{Y,t} + e_{t+1}) = 0. \tag{12}$$

On the one hand, the labor tax reduces consumption when young, which increases the marginal utility of young consumption and creates an incentive to save less (in order to smooth the reduction in young consumption).

On the other hand, the pension e_{t+1} increases consumption in old age, which decreases the marginal utility of old-age consumption and creates an incentive to reduce savings in order to consume more when young. Intuitively, the more generous pensions are (i.e., the higher e_{t+1} is) the lower the incentives to save in order to consume tomorrow.

Note that both forces reduce the incentives to save when young; therefore, PAYG systems have a negative impact on capital accumulation.

For the particular case of log utility, the FOC reads:

$$\frac{1}{w_t\,(1-\tau_{L,t})\, - s_{Y,t}} \;=\; \beta\,(1+r_{t+1})\frac{1}{(1+r_{t+1})s_{Y,t} + e_{t+1}},$$

which implies:[14]

$$s_{Y,t} \;=\; \frac{\beta}{1+\beta}\,w_t\,(1-\tau_{L,t})\; - \frac{e_{t+1}}{(1+\beta)(1+r_{t+1})}\,, \tag{13}$$

from which we confirm the previous intuitions: $s_{Y,t}$ decreases with $\tau_{L,t}$ and e_{t+1}. Further, recall that in the log utility case we had seen that the income and substitution effects offset each other, which is no longer true here: the substitution effect now dominates. Therefore, a higher interest rate induces consumers to save more when young due to consumption when old becoming cheaper.

[14] Derivation: first, we can rewrite it as follows: $\frac{1}{w_t\,(1-\tau_{L,t})\,-s_{Y,t}} \;=\; \beta\,\frac{1}{s_{Y,t}+\frac{e_{t+1}}{1+r_{t+1}}}$, which in turn implies $s_{Y,t} + \frac{e_{t+1}}{1+r_{t+1}} \;=\; \beta(w_t\,(1-\tau_{L,t})\; - s_{Y,t})$, and then $s_{Y,t}(1+\beta) \;=\; \beta\, w_t\,(1-\tau_{L,t})\; - \frac{e_{t+1}}{1+r_{t+1}}$.

The firm's problem: The firm's problem is exactly the same as before. The production function is $F(L_t, K_t) = A\, K_t^{\alpha}\, L_t^{1-\alpha}$, and in per capita terms: $\frac{F(K_t, L_t)}{L_t} = f(k_t) = A\, k_t^{\alpha}$. Then (as derived in section 1.3) the FOCs can be written as: $w_t = f(k_t) - f'(k_t)\, k_t$ and $r_{t+1} = f'(k_{t+1}) - \delta$; and so:

$$w_t = (1-\alpha)\, A\, k_t^{\alpha}\,, \qquad r_{t+1} = \alpha\, A\, k_{t+1}^{\alpha-1} - \delta\,. \tag{14}$$

Law of motion of capital: Using the savings function found in Equation 13, then equation 3 reads:

$$k_{t+1} \;=\; \frac{1}{n}\left[\frac{\beta}{1+\beta}\, w_t\,(1-\tau_{L,t}) \;-\; \frac{e_{t+1}}{(1+\beta)(1+r_{t+1})}\right]. \tag{15}$$

And using the factor prices w_t, r_{t+1} from equation 14 and e_{t+1} from the balanced budget rule in equation 11 (also, in order to solve for k_{t+1} in closed-form, let's suppose full depreciation after one period, that is $\delta = 1$):[15]

$$k_{t+1} \;=\; \frac{1}{n_{t+1}} \frac{\beta\, \alpha(1-\alpha) A\, k_t^{\alpha}\,(1-\tau_{L,t})}{(1+\beta)\alpha + (1-\alpha)\,\tau_{L,t+1}}. \tag{16}$$

From this equation, we see:

- Higher labor taxes today (i.e., $\uparrow \tau_{L,t}$) reduce capital accumulation. This is due to the channel discussed earlier: higher $\tau_{L,t}$ reduces consumption when young, leading to a reduction in savings in order to smooth consumption.

- Higher labor taxes tomorrow (i.e., $\uparrow \tau_{L,t+1}$) reduce capital accumulation today. This is because higher $\tau_{L,t+1}$ translates into higher pensions e_{t+1} (due to the government balanced budget condition) and, as we have already seen, the anticipation of future pensions in old age reduces the incentive to save for consumption in old age.

- Higher capital today (i.e., $\uparrow k_t$) leads to higher capital tomorrow. This is because higher capital today implies a higher marginal product of labor, which in turn implies higher wages. As discussed in Section 1.2.2, consumers translate an increase in wages when young into higher saving (recall the intuition: $\uparrow w_t$ means higher income when young, and since consumers want to smooth consumption, they increase both $c_{Y,t}$ and $s_{Y,t}$).

[15] When we substitute e_{t+1}, w_t, r_{t+1}, we get $k_{t+1} = \frac{1}{n_{t+1}} \left[\frac{\beta}{1+\beta}\,(1-\alpha)A k_t^{\alpha}\,(1-\tau_{L,t}) - \frac{n_{t+1}\,(1-\alpha)A\, k_{t+1}^{\alpha}\,\tau_{L,t+1}}{(1+\beta)\,\alpha\, A\, k_{t+1}^{\alpha-1}}\right]$, simplifying a bit: $k_{t+1} = \frac{\beta}{1+\beta}\,\frac{(1-\alpha)A}{n_{t+1}} k_t^{\alpha}\,(1-\tau_{L,t}) - k_{t+1}\,\frac{(1-\alpha)\,\tau_{L,t+1}}{(1+\beta)\,\alpha}$. And solving for k_{t+1}: $k_{t+1}\left(1 + \frac{(1-\alpha)\,\tau_{L,t+1}}{(1+\beta)\,\alpha}\right) = \frac{\beta}{1+\beta}\,\frac{(1-\alpha)A}{n_{t+1}} k_t^{\alpha}\,(1-\tau_{L,t})$, and multiplying both sides by $(1+\beta)\alpha$: $k_{t+1}\,((1+\beta)\alpha + (1-\alpha)\,\tau_{L,t+1}) = \beta\,\frac{\alpha(1-\alpha)A}{n_{t+1}} k_t^{\alpha}\,(1-\tau_{L,t})$.

- Higher population growth (i.e., $\uparrow n_{t+1}$) reduces capital per capita. Consumers' saving decisions are independent of population growth, so higher population in the next period implies that the same amount of capital is spread across a larger population.